



Research paper

A path planning method based on deep reinforcement learning for AUV in complex marine environment

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ABSTRACT

The potential of autonomous underwater vehicle (AUV) on future applications is significant due to advancements in autonomy and intelligence. Path planning is a critical technology for AUV to perform operational missions in complex marine environments. To this end, this paper proposes a path planning method for AUV based on deep reinforcement learning. Initially, considering actual requirements, a complex marine environment model containing underwater terrain, sonobuoy detection, and ocean currents is established. Subsequently, the corresponding state space, action space, and reward function are formulated. Furthermore, to address the inherent limitations of existing deep reinforcement learning algorithms in terms of training efficiency, a mixed experience replay (MER) strategy is proposed. This strategy aims to enhance the efficiency of sample learning by integrating prior knowledge and exploration experience. Lastly, a novel HMER-SAC algorithm for AUV path planning is proposed by integrating the Soft Actor–Critic (SAC) algorithm with the hierarchical reinforcement learning strategy and the MER strategy. The results of the simulation and experiment demonstrate that the method is capable of efficiently planning executable paths in complex marine environments and exhibits superior training efficiency, stability, and performance.

1. Introduction

In recent years, researchers have demonstrated a growing interest in autonomous underwater vehicle (AUV) due to the advancements in sensors, information technology, unmanned control, artificial intelligence, and related fields. Categorized as an underwater robot, AUVs demonstrate the capability to perform a wide range of operational tasks underwater, with applications in civil, scientific, and military domains (Zeng et al., 2015). In the context of military operations, AUVs offer several advantages over traditional manned submarines. Their smaller size and unmanned characteristics allow them to operate in a more covert manner, conducting reconnaissance, surveillance, combat, and other operations in complex marine environments (Bogue, 2015).

Path planning is one of the key technologies for AUV to fulfill the above operations autonomously, which has great impact on the effectiveness of the missions. The core objective of the path planning is to guide the AUV from the starting position to the target area along the optimal navigation path under certain conditions and constraints (Cheng et al., 2021). The key factors to be considered include submarine terrain collision constraints, motion performance constraints of the AUV, and uncertainty risks that may be encountered during navigation, such as enemy sonar detection and ocean current interference.

Research on AUV path planning has yielded numerous methodological outcomes, categorized based on application scenarios as global scenarios with known static obstacles and local scenarios featuring unknown and dynamic obstacles (Cheng et al., 2021). With regard to the time of execution, the methods can be divided into two categories: preliminary path planning, which is conducted before the commencement of the mission, and path replanning, which is carried out during the mission (Zeng et al., 2015). From a computational standpoint, these methods fall into several types, including geometric model search-based methods, virtual potential field-based methods, and evolutionary algorithm-based methods (Wang et al., 2023).

The geometric model search-based methods represent the most traditional approach, which obtain feasible paths by exploring the digital map in random, probability, or heuristic under the objective function, mainly including the Rapidly-exploring Random Trees (RRT) (Taheri et al., 2019; Meng et al., 2023; Feng et al., 2024) and A* algorithm (Chen et al., 2022; Zhang et al., 2023). These methods are suitable for simpler scenarios, yielding path results that align well with the objective function in a shorter time. However, due to the exponential explosion catastrophe, they are not applicable to complex scenarios. Virtual potential field-based methods direct the AUV towards

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the target area while avoiding obstacles through the establishment of a virtual potential field. Principal methods within this category include the Artificial Potential Field (APF) (Zhang et al., 2023; Yu and Ning, 2023) and Dynamic Velocity Potential Field (DVFD) (Zhao et al., 2024). These methods boast advantages in addressing real-time dynamic path planning problems, but there are dilemmas such as the local minimization problem and the inability to comprehensively simulate complex environments. The methods based on evolutionary algorithms refer to the use of Particle Swarm Optimization (PSO) (Lim et al., 2022; Zhi and Zuo, 2024), Genetic Algorithm (GA) (Hao et al., 2022; Tian et al., 2022), Whale Optimization Algorithm (WOA) (Yan et al., 2022; Huang et al., 2023), and other algorithms to realize path planning. These methods optimize the generated path continuously according to the objective function to obtain better solutions, which can be applied in complex environments. However, there are problems such as long computation time, high computational complexity, and large demand for computational resources.

The underwater environment in which AUV operates the missions is complex 3D scenarios including underwater terrain, ocean currents and other elements. To better approximate the real world, the AUV needs to have a strong local path planning capability in unknown dynamic environments. For the above methods, the geometric model search-based methods and virtual potential field-based methods cannot be well adapted to such complex environments with multiple elements due to the limitations of the algorithms. Although the methods based on evolutionary algorithms can be applied in complex environments, their computationally time-consuming and high resource requirements make them unsuitable for local path planning problems.

In recent years, with the advancement of artificial intelligence technology, researchers have shown increasing interest in utilizing deep reinforcement learning to solve the problem of AUV local path planning in unknown dynamic environments. After training, the deep reinforcement learning-based path planning methods can guide the AUV to make appropriate decisions according to the surrounding environment until reaching the target area. The output of these methods is typically used to select actions from the action space or directly control the AUV's rudder, elevator, and propellers (Wang et al., 2023). The aforementioned characteristics enable the path planning method based on deep reinforcement learning to address the AUV local path planning problem in unknown, complex, and dynamic marine environments (Cheng et al., 2021). Currently, various deep reinforcement learning algorithms, including Deep Deterministic Policy Gradient (DDPG), Proximal Policy Optimization (PPO), and Soft Actor Critic (SAC), have been tried and applied in different scenarios and tasks.

Lan et al. (2022) focuses on underwater gliders (UG) and their formation and introduces a DDPG-based path planning algorithm capable of autonomously navigating a single UG and coordinating UGs in a 2D simulation environment with dense obstacles and ocean currents. In Hadi et al. (2022), the adaptive path planning for AUVs in unknown 2D environments with obstacles and ocean currents is investigated using the twin-delayed deep deterministic policy (TD3) algorithm, an enhanced version of DDPG. The system is designed to operate without prior environmental knowledge and demonstrates resilience to disturbances caused by ocean currents. Xu et al. (2022) introduces an event-triggered Soft Actor-Critic algorithm for AUV collision avoidance. The algorithm is applied to unknown static obstacle environments and unknown dynamic obstacle environments in 2D/3D spaces. In Xi et al. (2022), the Double Dueling Deep Q-Network (D3QN) algorithm is employed to execute path planning for AUVs in complex environments that include 3D seafloor topographic space and local ocean current information. Gao et al. (2023) proposes a collision avoidance planning algorithm for unmanned underwater vehicle (UUV) based on PPO and the dynamic windows approach. The algorithm demonstrates the ability to efficiently plan smooth collision-free paths in complex obstacle environments within an acceptable time.

As discussed in the Wang et al. (2023), Cheng et al. (2021) and summarized from the studies above, the current utilization of deep reinforcement learning to solve the AUV path planning problem presents the following challenges:

- *Sample inefficiency*: In the absence of prior knowledge, relying solely on experience gained from agent–environment interactions will inevitably result in sample inefficiency, ultimately leading to suboptimal training efficiency and poor training outcomes.
- *High-dimensional state space*: As the complexity of simulation environments increases, the agent requires a greater quantity and diversity of information about its surroundings. The high-dimensional state space used to express this surrounding information makes the training process of deep reinforcement learning more time-consuming and less reliable.

In response to the aforementioned challenges, the main innovations and contributions of this paper are summarized as follows:

- To address sample inefficiency, a mixed experience replay strategy for deep reinforcement learning is proposed, which aims to make use of more valuable training samples by mixing prior knowledge and exploration experience.
- For the challenge of high-dimensional state spaces, we employ the hierarchical reinforcement learning strategy to decompose high-dimensional complex problems into multiple simpler subproblems. Each subproblem only requires the corresponding lower-dimensional sub-state space, effectively achieving dimensionality reduction.

In this paper, a path planning method based on SAC is proposed for the AUV local path planning problem in unknown complex marine environments. The novel method combines the hierarchical reinforcement learning strategy with a mixed experience replay strategy and can choose appropriate actions for the AUV through surrounding environmental information to reach the target position. Furthermore, detailed explanations of the process and implementation framework of the proposed method are provided.

The subsequent sections of this paper are organized as follows. The problem description, environmental modeling, and deep reinforcement learning are profiled in Section 2. Then, in Section 3, we describe the implementation details of the proposed method. In Section 4, we utilize the simulation platform to train, test, and analyze the proposed method. In the end, we summarize the main contributions of this paper and point out directions for future work in Section 5.

2. Problem formulation

2.1. Problem description

In military applications, the ability of the AUV to accurately, safely, and stealthily reach the predetermined target area has a direct impact on the effectiveness of the operational mission. In this paper, we take the AUV performing an underwater operational mission as an example to carry out the study.

The scenario for AUV underwater operational mission is illustrated in Fig. 1. The objective is for the AUV to navigate safely and quickly from the starting position to the target area through path planning without being discovered by the enemy. During the path planning process, the AUV must steer clear of collision hazards of the underwater terrain and the threat areas of the enemy sonobuoys. In addition, due to the size constraints of the AUV, its power and energy are limited, making it imperative to utilize ocean currents to minimize energy consumption during the path planning process.

In accordance with the AUV underwater operational mission scenario, the primary considerations for evaluating the AUV path planning results include path length, safety factor, and ocean currents utilization

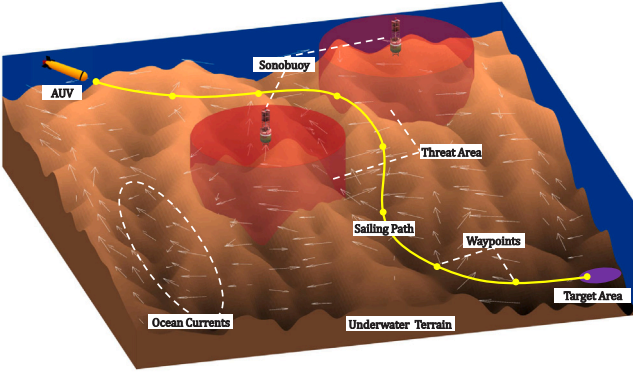


Fig. 1. Scenario for AUV underwater operational mission.

factor. The optimization model for the AUV path planning problem is defined as follows.

$$\begin{aligned} & \max F(f_d(\mathbf{P}), f_s(\mathbf{P}), f_e(\mathbf{P})) \\ & \text{s.t.} \begin{cases} p_1 = p_O \\ p_n = p_T \\ p_i \notin O_{\text{collision}}, \forall i \in [1, n] \end{cases} \end{aligned} \quad (1)$$

where the function $F(\cdot)$ denotes the comprehensive evaluation of the path obtained through planning, and the individual evaluations for the path length, safety factor and ocean currents utilization factor are represented by the $f_d(\mathbf{P})$, $f_s(\mathbf{P})$, and $f_e(\mathbf{P})$, respectively; p_O and p_T denote the start position and target position respectively; $\mathbf{P} = \{p_1, p_2, \dots, p_n\}$ denotes the path obtained by planning, which is a set of waypoints; $O_{\text{collision}}$ denotes the terrain collision constraints. The mathematical model means that the AUV path planning problem expects a solution with the shortest path length, optimal safety and optimal currents utilization.

2.2. Environmental modeling

The marine environment in the above scenario consists of a 3D space that includes the underwater terrain, the detection areas of enemy sonobuoys, and the ocean currents. We perform environmental modeling for the path planning problem based on the above mentioned aspects.

2.2.1. Underwater terrain modeling

To enhance the realism of the simulation, the underwater terrain is represented by digital elevation model (DEM). DEM simulates the terrain configuration using a finite set of terrain elevation data defined as follows:

$$\mathbf{G} = \left\{ \left(x_i, y_j, g(x_i, y_j) \right) \mid [x_i, y_j]^T \in \mathbf{D} \right\} \quad (2)$$

where $\mathbf{G}$ means the DEM database; $[x_i, y_j]^T \in \mathbf{D}$ represents the coordinate in the two-dimensional plane, and $g(x_i, y_j)$ signifies the terrain elevation data at the coordinate.

Because of the discrete points used to record terrain information at regular intervals, the DEM cannot represent the elevation information of the positions in the intervals. Interpolation calculation is necessary during AUV path planning to determine the elevation of non-existent positions in the DEM. Commonly employed methods of interpolation include Nearest Neighbor Interpolation, Bilinear Interpolation, and Bicubic Interpolation, among others (Zhang and Yu, 2022). To balance smooth interpolation with minimal computing resource requirements, the Bilinear Interpolation (Chang, 2008) approach is utilized to approximate elevation data for any point, shown in Fig. 2.

Assuming the position of p is given by the coordinate $[x, y]^T$ where $x_i < x < x_{i+1}$, $y_j < y < y_{j+1}$, and $g(x_i, y_j)$, $g(x_i, y_{j+1})$, $g(x_{i+1}, y_j)$, and

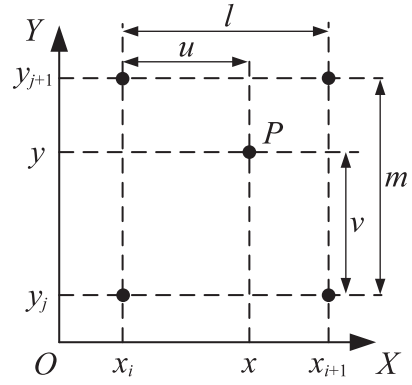


Fig. 2. Schematic diagram of Bilinear Interpolation.

$g(x_{i+1}, y_{j+1})$ are stored in the DEM database, the terrain elevation data $g(x, y)$ at position p can be calculated by Bilinear Interpolation as:

$$g(x, y) \approx \frac{1}{lm} \left[(l-u)(m-v) \cdot g(x_i, y_j) + uv \cdot g(x_{i+1}, y_{j+1}) + (l-u)v \cdot g(x_i, y_{j+1}) + u(m-v) \cdot g(x_{i+1}, y_j) \right] \quad (3)$$

where $l = x_{i+1} - x_i$ and $m = y_{j+1} - y_j$ denote the interval lengths of the DEM database along the x -axis and y -axis axes respectively; $u = x - x_i$ and $v = y - y_j$ represent the distances between position $p = [x, y]^T$ and position $[x_i, y_j]^T$ along the x -axis and y -axis respectively.

2.2.2. Sonobuoy detection modeling

Sonobuoys pose a significant threat to the performance of AUV operational missions. The AUV should minimize their presence in the detection areas to avoid discovery by sonobuoys. In reality, the sonar detection curve of the sonobuoy system is highly complex, exhibiting a non-monotonic variation in detection probability P_d per ensoufication with range (Fewell et al., 2008). The employment of high-fidelity acoustic propagation modeling to ascertain detection probability necessitates a more substantial investment of computing time and resources (Özer and Hocaoglu, 2021). It is therefore essential to employ a detection probability curve model that accurately reflects the most salient features of real detection probabilities while disregarding less pertinent details. This kind of model, which includes the cookie-cutter, exponential curve, and Fermi function, is predicated on the assumption that propagation loss primarily depends on distance. Consequently, the probability of detection is expected to decline as the target distance increases. In this paper, the Fermi function, the most prevalent in sonar analysis research and more aligned with realistic scenarios, is employed to establish the detection probability of a sonobuoy as follows (Özer and Hocaoglu, 2021).

$$P_d(R) = \frac{1}{1 + 10^{(R/R_0 - 1)/b}} \quad (4)$$

where $P_d(R)$ denotes the detection probability of range R ; R_0 is the range at which $P_d(R_0) = 0.5$; b represents the diffusivity parameter that regulates the gradient of the function at $P_d(R_0)$. Fig. 3 illustrates the curve of detection probability with range for sonobuoy when $R_0 = 1500$ meters and $b = 0.5$.

2.2.3. Ocean currents modeling

Ocean currents significantly impact the movement of AUV. The generation of currents is a complex phenomenon influenced by a multitude of factors, including wind, the Coriolis effect, alterations in seabed topography, the distribution of land and sea, and other variables. In the present research, ocean currents models can be acquired via different methods, primarily classified into sensing equipment detection (e.g., moored currents meter, acoustic Doppler profiler, high-frequency radar, etc.) and numerical fitting models.

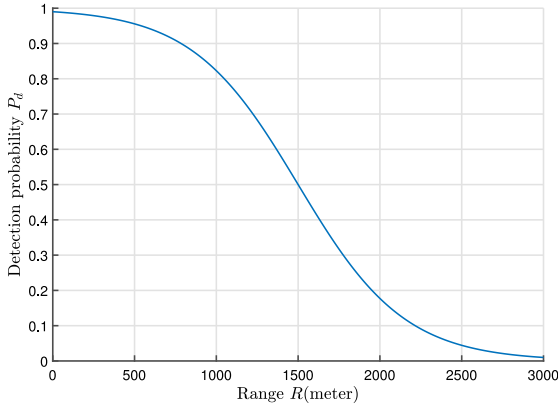


Fig. 3. Detection probability versus range curve based on the Fermi function.

In this paper, we utilize the numerical solution of the Navier–Stokes equation for ocean currents, employing the viscous Lamb vortex model (Cai et al., 2021; Zadeh et al., 2017). This ocean currents model constructs 3D turbulent currents through the integration of multiple layered 2D currents, which are defined as Eq. (5).

$$\mathbf{V}_c = [v_{cx}, v_{cy}, v_{cz}]^T = f_c(\mathbf{p}_c, \Gamma_c, \delta_c) \quad (5)$$

$$\begin{cases} v_{cx}(\mathbf{p}) = -\Gamma_c \frac{y - y_c}{2\pi \cdot (\mathbf{p} - \mathbf{p}_c)^2} \left[1 - e^{-\frac{(\mathbf{p} - \mathbf{p}_c)^2}{\delta_c^2}} \right] \\ v_{cy}(\mathbf{p}) = \Gamma_c \frac{x - x_c}{2\pi \cdot (\mathbf{p} - \mathbf{p}_c)^2} \left[1 - e^{-\frac{(\mathbf{p} - \mathbf{p}_c)^2}{\delta_c^2}} \right] \\ v_{cz}(\mathbf{p}) = \gamma_c \sqrt{v_{cx}(\mathbf{p})^2 + v_{cy}(\mathbf{p})^2} \end{cases}$$

where $v_{cx}(\mathbf{p})$, $v_{cy}(\mathbf{p})$, and $v_{cz}(\mathbf{p})$ are the velocity components of currents in x -axis, y -axis and z -axis directions at the position $\mathbf{p} = [x, y, z]^T$ respectively; $\mathbf{p}_c = [x_c, y_c, z_c]^T$, Γ_c , and δ_c denote the coordinates, strength, and radius of the vortex respectively; γ_c is a proportional constant that reflects weak vertical motions of the ocean currents.

In general, it is possible that a specific region of the ocean may contain multiple vortices that influence the overall ocean currents. In such a case, the ocean current at a given location within the region can be conceptualized as the vector sum of the currents generated by all vortices, as defined by the following.

$$\mathbf{V}_c = [v_{cx}, v_{cy}, v_{cz}]^T = \left[\sum_{i=1}^{N_c} v_{cx,i}, \sum_{i=1}^{N_c} v_{cy,i}, \sum_{i=1}^{N_c} v_{cz,i} \right]^T \quad (6)$$

where N_c is the number of the vortices.

2.3. Deep reinforcement learning

Reinforcement learning is a method of training an agent through environmental interactions to develop an effective operational policy based on perception of the surrounding environment. Markov Decision Process (MDP) is a basic framework for reinforcement learning, defined the agent–environment interaction as a quintuplet containing the state space $\mathcal{S}$, action space $\mathcal{A}$, the state transition probability P , reward function R and as discount factor γ as

$$\mathbf{M} = \langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle \quad (7)$$

where $\mathcal{S}$ is the set of possible states; $\mathcal{A}$ is the set of actions for agent; $P(s_{t+1}|s_t, a_t) \in [0, 1]$ represents the state transition probability when the agent takes the action a_t from state s_t to s_{t+1} ; R is the reward obtained from interaction; the discount factor $\gamma \in [0, 1]$ balances the importance of current with future rewards.

In reinforcement learning, the agent follows policy π to take action a_t in state s_t at each time-step t . The policy π is a mapping from the

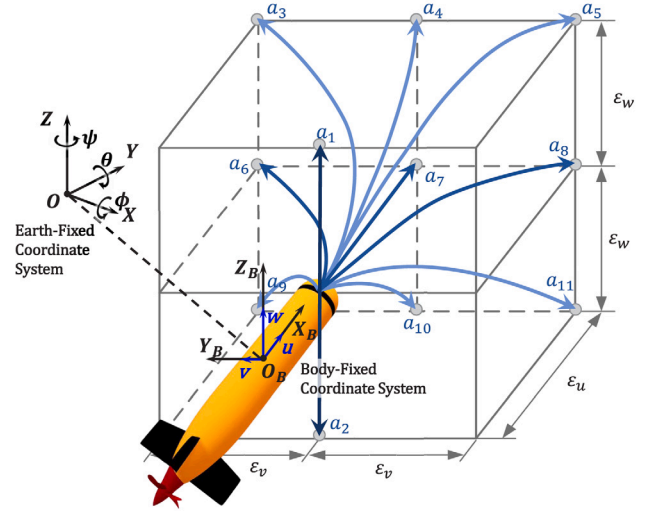


Fig. 4. Schematic diagram of the AUV motion model and action space.

state space to a probability distribution over the action space as $\pi : \mathcal{S} \rightarrow \mathcal{A}$. During the train process, reinforcement learning continuously optimizes the policy π by maximizing the expectation of the decay sum of rewards. Generally, the action-state value function $Q^\pi(s_t, a_t)$, also called Q-value, is used to represent the above expectation as

$$Q^\pi(s_t, a_t) = \mathbb{E}_{a_t \sim \pi} [R + \gamma Q^\pi(s_{t+1}, a_{t+1})] \quad (8)$$

and the optimal policy is defined as

$$\pi^*(s_t) = \arg \max_{\pi} Q^{\pi^*}(s_t, a_t) \quad (9)$$

Deep reinforcement learning combines reinforcement learning with the powerful fitting capabilities of deep neural networks. This approach leverages deep neural networks to represent the value function, addressing the curse of dimensionality in reinforcement learning. In recent years, several deep reinforcement learning methods have been developed and applied in various fields, including path planning and autonomous navigation.

3. Methodology

3.1. MDP for AUV path planning

This subsection provides a description of the MDP model for the AUV path planning problem, including action space, state space, and reward functions.

3.1.1. AUV motion model and action space

In this paper, we describe the position and attitude of AUV by a motion model, defining earth-fixed coordinate system and the body-fixed coordinate system and shown in Fig. 4. The earth-fixed coordinate system is coupled to the earth, with the X -axis pointing north, the Y -axis pointing west and the Z -axis pointing vertically upward. The body-fixed coordinate system is coupled to the AUV, where the X_B -axis, Y_B -axis, and Z_B -axis point to the surge direction, sway direction, and heave direction respectively.

The position and Euler angles of the AUV are defined as $\mathbf{p} = [x, y, z]^T$ and $\boldsymbol{\omega} = [\phi, \theta, \psi]^T$ on the earth-fixed coordinate system respectively. The linear velocity of AUV is defined as $\mathbf{v} = [u, v, w]^T$ on the body-fixed coordinate system. Considering the effect of ocean currents and ignoring the influence of lateral and roll movement, the kinematic

model of the AUV in ocean currents is defined as Eq. (10) (Xu et al., 2022).

$$\dot{\mathbf{p}} = \mathbf{T}(\theta, \psi) \mathbf{v} + \mathbf{V}_c \quad (10)$$

where the transformation matrix $\mathbf{T}(\theta, \psi)$ is defined as

$$\mathbf{T}(\theta, \psi) = \begin{bmatrix} \cos \theta \cos \psi & -\sin \psi & \sin \theta \cos \psi \\ \cos \theta \sin \psi & \cos \psi & \sin \theta \sin \psi \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \quad (11)$$

and then, the Euler angles can be calculated as

$$\begin{cases} \psi = \arctan\left(\frac{v_y}{v_x}\right) \\ \theta = \arctan\left(\frac{v_z}{\sqrt{v_x^2 + v_y^2}}\right) \end{cases} \quad (12)$$

Following the above AUV model, the action space is constituted by a set of all feasible actions that the AUV can perform for path planning. The action space designed in this paper contains 11 different motion actions as Eq. (13) and the spatial relationship between each action is shown in Fig. 4.

$$\begin{cases} \mathbf{a} = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{11}\} \\ \mathbf{a}_i = [u_i, v_i, w_i]^T = [k_{u,i}\epsilon_u, k_{v,i}\epsilon_v, k_{w,i}\epsilon_w]^T \end{cases} \quad (13)$$

where u_i , v_i , and w_i are linear velocities of the action $\mathbf{a}_i$ on the body-fixed coordinate system; ϵ_u , ϵ_v , and ϵ_w are the values for each velocity; $k_{u,i}$, $k_{v,i}$, $k_{w,i} \in \{1, -1, 0\}$ are constants for $\mathbf{a}_i$ to control the velocities to be forward, reverse or none. For example, the action $\mathbf{a}_{11}$ can be defined as $\mathbf{a}_{11} = [\epsilon_u, -\epsilon_v, -\epsilon_w]^T$ where $k_{u,11} = 1$, $k_{v,11} = -1$, $k_{w,11} = -1$.

In this action space, $\mathbf{a}_1$ and $\mathbf{a}_2$ indicating upward floating and downward sinking, and $\mathbf{a}_6$, $\mathbf{a}_7$, and $\mathbf{a}_8$ indicating movement to the left front, direct front, and right front, and the other actions can be regarded as the sum of two above actions. When the AUV performs an action $\mathbf{a}_i$, its position state will be transformed as

$$\mathbf{p}_{t+1} = \mathbf{p}_t + [\mathbf{T}(\theta, \psi) \mathbf{a}_i + \mathbf{V}_c] \Delta t \quad (14)$$

3.1.2. State space

During the algorithm training and application process, the AUV is required to evaluate the current situation and take appropriate action through the state information received from the environment. Therefore, designing a rational state space is crucial. It is typically necessary to adhere to three guidelines for state space design: task-oriented, orthogonalization, and normalization (Xu et al., 2022).

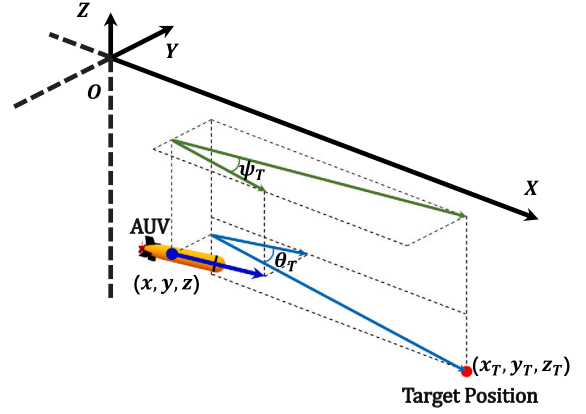
For the AUV path planning problem, the AUV requires not only target position but also environmental information such as obstacle information from the surrounding terrain, threat information from sonobuoys, and ocean currents information. Following the above criteria, the state space designed in this paper consists of the following parts in order to fully reflect the various state information of the AUV.

(1) Relative to the target position

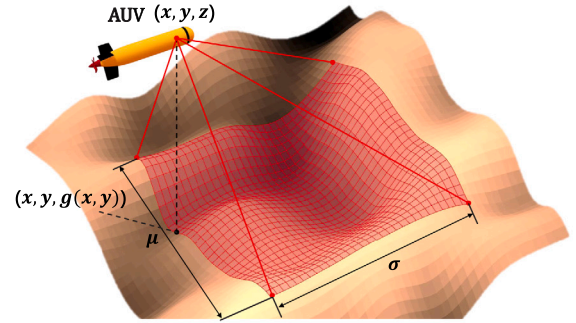
The primary objective of the AUV path planning problem is to arrive at the target position. Therefore, it is necessary to provide relative position and angle information of the AUV to the target position, which is defined as follows.

$$\mathbf{s}_1 = \left[\frac{x - x_T}{|x_O - x_T|}, \frac{y - y_T}{|y_O - y_T|}, \frac{z - z_T}{|z_O - z_T|}, \frac{\theta_T}{\pi}, \frac{\psi_T}{\pi} \right] \quad (15)$$

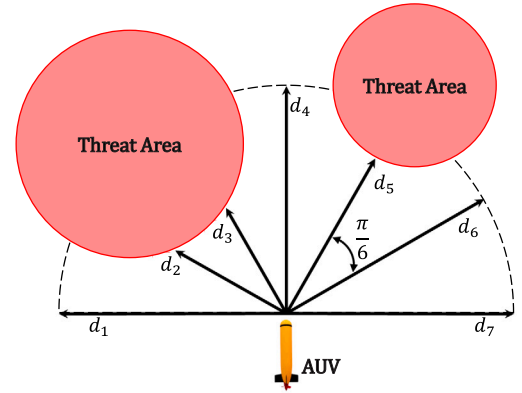
where $\mathbf{p} = [x, y, z]^T$, $\mathbf{p}_O = [x_O, y_O, z_O]^T$, and $\mathbf{p}_T = [x_T, y_T, z_T]^T$ denote the current, start and target positions of the AUV respectively, θ_T and ψ_T respectively represent the pitch angle difference and heading angle difference between the AUV and the target position, as shown in Fig. 5(a).



(a) State space relative to the target position



(b) State space relative to the surrounding terrain



(c) State space relative to the threat areas

Fig. 5. Schematic diagram of the state space.

(2) Relative to the surrounding terrain

During operational missions, the AUV needs to keep as close to the seafloor as possible in order to capitalize on changes in terrain and avoid detection by the enemy. Therefore, the AUV requires the capability to obtain surrounding terrain information for obstacle avoidance. In the configuration of this paper, the AUV can sense changes in the terrain ahead through terrain scanning equipment, in which the forward detectable range of the terrain scanning device is σ and the lateral detection range is μ , as shown in Fig. 5(b). The AUV acquires the state space relative to the surrounding terrain from the sensing devices, defined as follows.

$$\mathbf{s}_2 = \frac{1}{k_G} \text{Flatten}(z \cdot O_{[N_\sigma \times N_\mu]} - G_{[N_\sigma \times N_\mu]}) \quad (16)$$

where Flatten($\cdot$) denotes the function to transform a matrix into a vector, $G_{[N_\sigma \times N_\mu]}$ denotes the terrain information matrix containing DEM data with N_σ rows and N_μ columns, $O_{[N_\sigma \times N_\mu]}$ denotes a matrix of ones with the same size of G , k_G is a normalized constant.

(3) Relative to the threat areas

The detection range of the sonobuoy is typically larger in the horizontal direction than in the vertical direction, so the threat area can be approximated as a cylinder, which makes the AUV path planning problem of threat avoidance downgraded from the three-dimensional to the two-dimensional. Drawing on Zhang et al. (2022), the state space relative to the threat area is shown in Fig. 5(c) and defined as follows.

$$s_3 = \left[\frac{d_1}{d_{\max}}, \frac{d_2}{d_{\max}}, \dots, \frac{d_7}{d_{\max}} \right] \quad (17)$$

where d_i denotes the distance from the AUV to the threat area, the angle between d_i and d_{i+1} is $\pi/6$, $d_{\max}$ denotes the perception radius.

(4) Relative to the ocean currents

The state space relative to the ocean currents is defined as

$$s_4 = \left[\frac{u_c}{k_u}, \frac{v_c}{k_v}, \frac{w_c}{k_w} \right] \quad (18)$$

where u_c , v_c , and w_c denote the velocity components of the ocean current at the position of the AUV, k_u , k_v , and k_w denote the normalized constants for the components respectively.

As a conclusion, the final state space is formulated as Eq. (19) by combining the previous parts.

$$s = [s_1, s_2, s_3, s_4] \quad (19)$$

3.1.3. Reward function

During the process of solving the path planning problem by deep reinforcement learning, the reward function is designed to help the AUV perform the desired action and achieve the set task objective. In order to determine the optimal path for the AUV to reach the target position while avoiding terrain collisions and detection by enemy sonobuoys, we set a series of sub-reward functions based on the evaluation indexes in Eq. (1). In these sub-reward functions, the distance reward is employed to represent the evaluation of path length $f_d(\mathbf{P})$, the collision avoidance reward and the detection avoidance reward are utilized to represent the evaluation of safety factor $f_s(\mathbf{P})$, and the ocean currents utilization reward is applied to represent the evaluation of ocean currents utilization factor $f_e(\mathbf{P})$.

(1) Distance reward

During path planning, it is essential for the AUV to approach the target position gradually. This entails providing rewards for actions that bring the AUV closer to the target position, with the magnitude of the reward increasing as the distance between the AUV and the target decreases. The sub-reward function for decreasing distance is defined as follows.

$$r_1 = -D/D_0 \quad (20)$$

where D_0 and D denote the Euclidean distance between the AUV and the target position at previous and current time-step respectively.

(2) Collision avoidance reward

During navigation, the AUV must maintain constant vigilance to avoid collisions with topographic obstacles present on the seafloor. Simultaneously, the AUV needs to leverage the masking effect offered by the seabed and navigate close to it, minimizing the probability of detection by unknown sonars. As a result, the sub-reward function for collision avoidance is defined as follows.

$$r_2 = \begin{cases} \sin(\frac{\pi}{H_s} \Delta_z - \frac{\pi}{2}) & \text{if } 0 < \Delta_z \leq H_s \\ \sin(\frac{\pi}{2H_s} \Delta_z) & \text{if } H_s < \Delta_z \leq 2H_s \\ e^{(1 - \frac{\Delta_z}{2H_s})\pi} - 1 & \text{if } \Delta_z > 2H_s \end{cases} \quad (21)$$

where $\Delta_z = z - g(x, y)$, $g(x, y)$ denotes the seafloor topographic elevation at the position of the AUV calculated by Eq. (3); H_s denotes the safe distance to the seafloor.

In addition, if the AUV collides with an obstacle on the terrain as $\Delta_z \leq 0$, the mission is considered as failed, and the AUV receives the appropriate terminal reward, as detailed later.

(3) Detection avoidance reward

The AUV needs to arrive at the target area stealthily during operational missions, which requires avoiding detection by enemy sonobuoys. Combined with the sonobuoy detection model constructed in Section 2.2.2, we set that the higher the probability of AUV not being detected, the higher the reward value obtained in this paper. The sub-reward function for detection avoidance is defined as follows.

$$r_3 = \prod_i^{N_d} (1 - P_{d_i}) \quad (22)$$

where N_d denotes the number of enemy sonobuoys; P_{d_i} denotes the probability of the current position of the AUV being detected by a sonobuoy, calculated through Eq. (4).

(4) Ocean currents utilization reward

In a non-emergency state, it is recommended to minimize the energy consumption of the AUV by utilizing the forceful push of ocean currents. In this paper, we establish that a smaller angle between the AUV heading direction and ocean current velocity direction leads to a higher reward value. The sub-reward function for ocean currents utilization is defined as follows.

$$r_4 = \cos(\Delta\psi) \quad (23)$$

where $\Delta\psi$ denotes the angle between the AUV heading direction and ocean current velocity direction.

(5) Terminal reward

Additionally, the AUV requires feedback on mission success or failure. The mission is considered successful when the AUV reaches the target position without collision or detection. Conversely, the mission is classified as a failure if a collision occurs or if the AUV is out of range of the mission area, or if it is detected by any of the sonobuoys with a detection probability greater than 50%. The AUV acquired reward of r_s or r_f for mission succeeds or fails respectively.

In summary, the complete reward function is eventually defined as follows.

$$r = \begin{cases} r_s & \text{if succeeds} \\ r_f & \text{if fails} \\ \lambda_1 r_1 + \lambda_2 r_2 + \lambda_3 r_3 + \lambda_4 r_4 & \text{each time-step} \end{cases} \quad (24)$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are the weights of the sub-reward functions.

3.2. Path planning method based on deep reinforcement learning

3.2.1. SAC algorithm

The Soft Actor-Critic is an off-policy actor-critic reinforcement learning based on the maximum entropy proposed by Haarnoja et al. (2018). The main feature of SAC is the entropy regularization applied to policy optimization and the Q-value in Eq. (8) is rewritten as

$$Q^\pi(s_t, a_t) = \mathbb{E}_{a_t \sim \pi} [R + \gamma Q^\pi(s_{t+1}, a_{t+1}) + \alpha H(\pi(\cdot|s_t))] \quad (25)$$

where α is the temperature parameter to balance the policy entropy and reward; $H(\pi(\cdot|s_t))$ is the entropy of the policy π defined as

$$H(\pi(\cdot|s_t)) = -\log \pi(a_t|s_t) \quad (26)$$

According to the action space in Section 3.1.1, the method proposed in this paper is based on the SAC for discrete action settings(SAC-Discrete) (Christodoulou, 2019). The structure of the SAC-Discrete is

Algorithm 1 SAC-Discrete Algorithm

```

1: Randomly initialize policy network  $\pi_\phi$ , action-state value networks  $Q_{\theta_1}, Q_{\theta_2}$  with hyperparameters  $\phi, \theta_1, \theta_2$ 
2: Initialize target networks  $Q_{\bar{\theta}_1}, Q_{\bar{\theta}_2}$ :  $\bar{\theta}_1 \leftarrow \theta_1, \bar{\theta}_2 \leftarrow \theta_2$ 
3: Initialize replay buffer:  $B \leftarrow \emptyset$ 
4: repeat
5:   for each environment step do
6:     Observe state  $s_t$  and select action  $a_t \sim \pi_\phi(s_t)$ 
7:     Execute action  $a_t$  and observe next state  $s_{t+1}$ , reward  $R$ , and done signal  $d$ 
8:     Store the transition in replay buffer:  $B \leftarrow B \cup \{(s_t, a_t, R_t, s_{t+1}, d_t)\}$ 
9:   end for
10:  for each training step do
11:    Randomly sample a batch of transitions from replay buffer:  $B = \{(\tilde{s}_i, \tilde{a}_i, \tilde{R}_i, \tilde{s}_{i+1}, \tilde{d}_i) | i = 1, 2, \dots, N\}$ 
12:    Update soft Q-networks  $Q_{\theta_i}$  by:  $\theta_i \leftarrow \theta_i - \lambda_{\theta_i} \hat{\nabla}_{\theta_i} J_{Q_{\theta_i}}(\theta_i)$  for  $i \in \{1, 2\}$ 
13:    Update policy network  $\pi_\phi$  by:  $\phi \leftarrow \phi - \lambda_{\phi} \hat{\nabla}_{\phi} J_{\pi}(\phi)$ 
14:    Update temperature parameter  $\alpha$  by:  $\alpha \leftarrow \alpha - \lambda_{\alpha} \hat{\nabla}_{\alpha} J(\alpha)$ 
15:    Update target networks  $Q_{\bar{\theta}_i}$  by:  $\bar{\theta}_i \leftarrow \tau \theta_i + (1 - \tau) \bar{\theta}_i$  for  $i \in \{1, 2\}$ 
16:  end for
17: until convergence

```

shown in Fig. 6 which involves five networks: a policy network π_ϕ , two soft Q-networks Q_θ and two target soft Q-networks $Q_{\bar{\theta}}$, where ϕ, θ and $\bar{\theta}$ are the hyperparameters for the networks respectively. Besides, the replay buffer stores the transition $(s_t, a_t, R_t, s_{t+1}, d_t)$, called experience, for training the networks.

In agent–environment interaction, the agent chooses action obeying distribution probability from the policy network π_ϕ . The policy network updates the hyperparameters according to the experience obtained from the sample batch and the action-state value from the soft Q-networks. The loss function for the policy network is defined as follows.

$$J_{\pi}(\phi) = \mathbb{E}_{s_t \sim B} [\pi_\phi(s_t)^T (\alpha \log(\pi_\phi(s_t)) - \min_{i=1,2} Q_{\theta_i}(s_t))] \quad (27)$$

where B is a sample batch from replay buffer.

The SAC-Discrete utilizes two soft Q-networks with the same structure to avoid overestimation bias of Q-value. Furthermore, two target soft Q-networks are used to stabilize training and enhance convergence. The loss function for soft Q-networks is defined as

$$J_Q(\theta_i) = \mathbb{E}_{(s_t, a_t, R_t, s_{t+1}, d_t) \sim B} \left[\frac{1}{2} \left(Q_{\theta_i}(s_t) - (R_t + \gamma(1 - d_t) \min_{i=1,2} Q_{\bar{\theta}_i}(s_{t+1}) - \alpha \log(\pi_\phi(s_{t+1}))) \right)^2 \right] \quad (28)$$

where d_t is done signal, $d_t = 1$ for task completion and $d_t = 0$ otherwise. The hyperparameters of the target soft Q-networks are updated by soft update rule as

$$\bar{\theta}_i \leftarrow \tau \theta_i + (1 - \tau) \bar{\theta}_i \quad (29)$$

where τ is soft update rate.

In addition, the temperature parameter α is set as a trainable hyperparameter and the loss function is defined as

$$J(\alpha) = \pi_\phi(s_t)^T (-\alpha \log(\pi_\phi(s_t)) + \bar{H}) \quad (30)$$

where $\bar{H}$ is the desired minimum expected entropy.

Algorithm 1 shows the process of agent–environment interaction and the networks training process for the SAC-Discrete algorithm.

3.2.2. Mixed experience replay

During the deep reinforcement learning training process, the agent encounters several difficulties including low-quality actions in the initial phase of training, poor training efficiency, delayed feedback, and slow convergence. Inspired by previous researches (Schaul et al., 2015; Zhou et al., 2021; Wan et al., 2022), this paper proposes a Mixed Experience Replay(MER) strategy, which aims to encourage the agent to get through the initial inefficient training as quickly as possible by learning the prior knowledge obtained from the existing algorithms.

There are two experience replay buffers present in MER, namely the prior experience buffer B_P and the exploration experience buffer B_E , defined as follows.

$$B_{MER} = B_P \cup B_E \quad (31)$$

The prior experience buffer B_P stores knowledge acquired through existing algorithms. Besides, the exploration experience buffer B_E stores the experience from agent–environment interaction, similar to replay buffer in Fig. 6.

The experiences stored in the two buffers exhibit distinctive features for the training process of the agent. Deep reinforcement learning aims to maximize cumulative rewards during agent training, which differs from existing algorithms used to gain prior experiences. This means that prior experiences may not be sufficient for the agent that receive ongoing feedback during long-term exploration, but may be more appropriate for developing feasible action strategies during initial training. On the contrary, the experiences from the initial exploration of the agent–environment interaction bring low-quality actions. Still, progress in training enables the agent to accumulate increasingly valuable experiences through ongoing exploration, ultimately achieving the reinforcement learning objective.

To assist the agent in acquiring appropriate experiences at advisable moments, MER sample batch strategy manipulates the proportionality parameter ξ to generates a combination of mixed experiences, shown in Algorithm 2. During the initial training phase, sampling primarily acquires training sample batches from B_P . As training advances, the parameter ξ will be updated according to Eq. (32) to decrease the proportion of prior experiences in the training samples, allowing the agent to gain more exploration experience.

$$\xi = \frac{1}{\ln(e + k_{\xi} i_{\xi} + N_{\xi})} \quad (32)$$

where e is the Euler's number; i_{ξ} denotes the sampling times; k_{ξ} is a constant to control the decrease rate of the proportionality parameter ξ ; N_{ξ} is a Gaussian normal distribution to increase randomness during network training.

3.2.3. Hierarchical MER-SAC for AUV path planning

The scenarios for an underwater operational mission are in a multi-faceted environment with diverse influence elements, including underwater terrain, sonobuoys, and ocean currents. The complexity of these scenarios imposes limitations on the AUV path planning process, adding complexity to problem-solving. Existing methods using deep reinforcement learning often simplify the environment by reducing the number of influencing elements to tackle AUV path planning, which contradicts practical requirements. To this end, this paper draws inspiration from

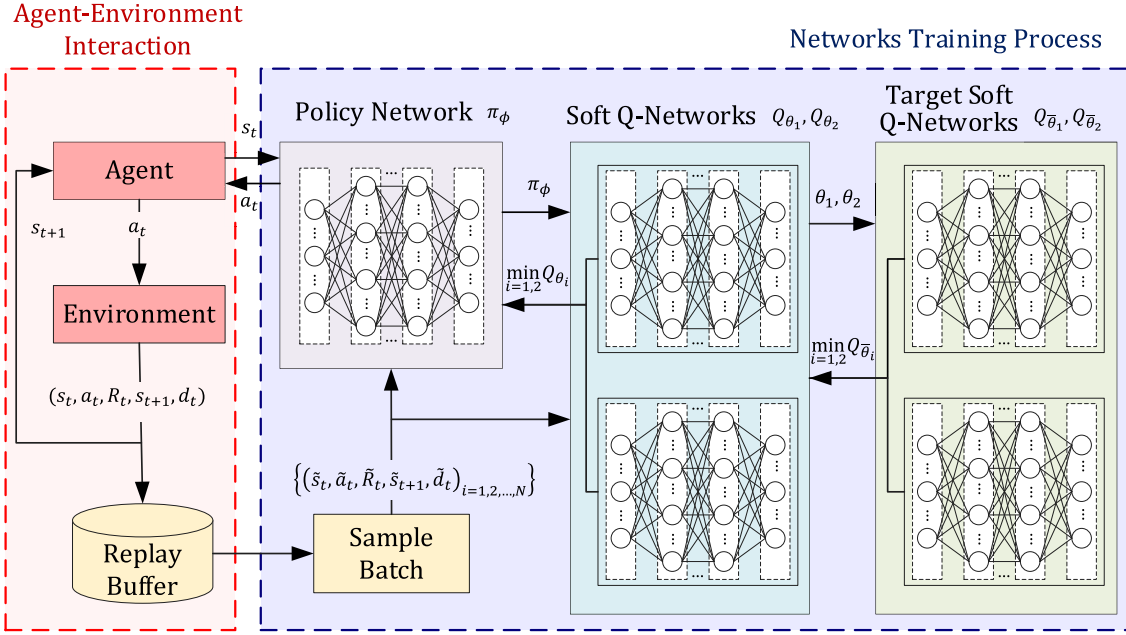


Fig. 6. Structure of the SAC-Discrete Algorithm.

Algorithm 2 MER sample batch strategy

```

Initialize proportionality parameter  $\xi = 1$ 
Initialize sampling times  $i_\xi = 0$ 
1: function SampleBatchMER( $N_B, B_P, B_E$ )
2:    $N_{B_P} = \text{ceil}(N_B \cdot \xi)$ 
3:    $N_{B_E} = N_B - N_{B_P}$ 
4:   Randomly sample a batch of  $N_{B_P}$  from  $B_P$ :  $b_P = \{(\tilde{s}_t^P, \tilde{a}_t^P, \tilde{R}_t^P, \tilde{s}_{t+1}^P, \tilde{d}_t^P) | i = 1, 2, \dots, N_{B_P}\}$ 
5:   Randomly sample a batch of  $N_{B_E}$  from  $B_E$ :  $b_E = \{(\tilde{s}_t^E, \tilde{a}_t^E, \tilde{R}_t^E, \tilde{s}_{t+1}^E, \tilde{d}_t^E) | i = 1, 2, \dots, N_{B_E}\}$ 
6:   Combine two batches as a MER batch:  $b_{MER} = b_P \cup b_E$ 
7:    $i_\xi \leftarrow i_\xi + 1$ 
8:   Update the proportionality parameter  $\xi$  by:  $\xi \leftarrow 1 / \ln(e + k_\xi i_\xi + N_\xi)$ 
9: return  $b_{MER}$ 

```

Algorithm 3 HMER-SAC for AUV path planning

```

1: Initialize Suggester Networks: Collision Avoidance Network, Detection Avoidance Network, Ocean Currents Utilization Network
2: Initialize Decision-Maker Network
3: Collect and store prior experiences to the MER of each network
4: repeat
5:   for each environment step do
6:     Observe state  $s_t$  and disseminate the corresponding part to each network by Data Distributor
7:     Obtain the action probability matrix  $P(a)$  from Suggester Networks
8:     Obtain the decision action  $a_D$  from Decision-Maker Network
9:     Select action  $a_t = \arg \max_{a_i \in a} [P(a) \cdot a_D]$ 
10:    Execute action  $a_t$  and observe next state  $s_{t+1}$ , reward  $R_t$ , and done signal  $d_t$ 
11:    Disseminate the transition to each network by Data Distributor
12:    Store the transition to the MER of each network
13:  end for
14:  if is training then
15:    for each training step do
16:      for each network do
17:        Sample a batch of transitions from MER by Algorithm 2
18:        Update SAC network by Step 12 to Step 15 in Algorithm 1
19:      end for
20:    end for
21:  end if
22: until convergence

```

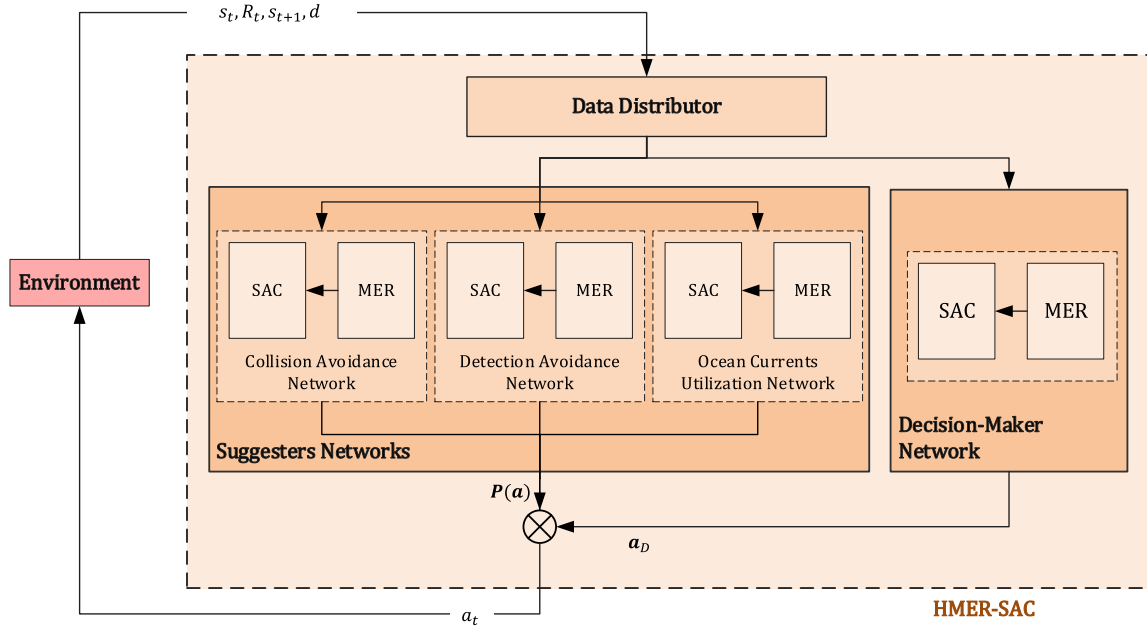


Fig. 7. Structure of the Hierarchical MER-SAC for AUV path planning.

the hierarchical reinforcement learning (HRL) strategy (Pope et al., 2021; Pateria et al., 2021) and proposes a method called HMER-SAC that aims to enable AUV path planning in complex environments.

The fundamental mechanism of HRL involves breaking down a highly complex problem into several simpler subproblems. The agent accomplishes the intricate task by addressing these subtasks. This approach delineates policies into low-level policies, which are tailored for specific subtasks and applied solely to achieve the corresponding objectives, and high-level policy, which assesses the comprehensive state and determines the ultimate action based on the low-level policies.

The proposed AUV path planning method based on HMER-SAC is depicted in Fig. 7. This method involves low-level policy networks named Suggester Networks for addressing subproblems, and a high-level policy network named Decision-Maker Network for making final policy. Within the Suggester Networks, the AUV path planning problem in complex environments is decomposed into three distinct subproblems: collision avoidance, detection avoidance, and ocean currents utilization. These subproblems are addressed through corresponding networks. All Suggester Networks and the Decision-Maker Network are developed by integrating the SAC algorithm and the MER strategy.

The state and reward information gathered by the agent during its interaction with the environment initially enters the Data Distributor. Since the training objectives of each network are different, the required state and reward information are also different. The Data Distributor is responsible for disseminating the necessary state and reward information to each of the Suggester Networks and the Decision-Maker Network.

The Collision Avoidance Network aims to prevent terrain collisions during navigation. Accordingly, the state and reward function for each time-step are designated as $s_{CA} = [s_1, s_2]$ and $r_{CA} = \lambda_1 r_1 + \lambda_2 r_2$. Similarly, the Detection Avoidance Network focuses on avoiding sonobuoy detection during navigation, with the state and reward represented as $s_{DA} = [s_1, s_3]$ and $r_{DA} = \lambda_1 r_1 + \lambda_3 r_3$. The objective of the Ocean Currents Utilization Network is to leverage the impetus of ocean currents in navigation. Its state and reward for each time-step are articulated as $s_{OU} = [s_1, s_4]$ and $r_{OU} = \lambda_1 r_1 + \lambda_4 r_4$. Lastly, the Decision-Maker Network necessitates global state and reward as s in Eq. (19) and r in Eq. (24).

Suggester Networks generate vectors comprising action probabilities for each subproblem requirement derived from the acquired local state

information, as depicted in Eq. (33).

$$\begin{cases} P_{CA}(a) = [P_{CA}(a_1), P_{CA}(a_2), \dots, P_{CA}(a_{11})]^T \\ P_{DA}(a) = [P_{DA}(a_1), P_{DA}(a_2), \dots, P_{DA}(a_{11})]^T \\ P_{OU}(a) = [P_{OU}(a_1), P_{OU}(a_2), \dots, P_{OU}(a_{11})]^T \end{cases} \quad (33)$$

where $P_{CA}(a)$, $P_{DA}(a)$, and $P_{OU}(a)$ are the action probability vectors from Collision Avoidance Network, Detection Avoidance Network, and Ocean Currents Utilization Network.

Subsequently, the action probability vectors from each Suggester Network are combined to construct an action probability matrix $P(a)$, depicted in Eq. (34).

$$P(a) = [P_{CA}(a), P_{DA}(a), P_{OU}(a)] \quad (34)$$

The Decision-Maker Network produces the ultimate decision action $a_D = [a_{CA}, a_{DA}, a_{OU}]^T$ for the subproblems determined by the global state, where a_{CA} , a_{DA} , and a_{OU} represent the current significance weighting of the three subproblems.

After the above process, the HMER-SAC method determines the final action a_t as follows.

$$a_t = \arg \max_{a_t \in a} [P(a) \cdot a_D] \quad (35)$$

Furthermore, to facilitate the operation of the MER before the training process, it is essential to inject prior knowledge into the MER of each network. To meet the specific requirements of each subproblem, distinct sources of prior knowledge are utilized. For all the Suggester Networks, we integrate prior knowledge from an improved A* algorithm (Zhang et al., 2016). As for the Decision-Maker Network, expert empirical knowledge from AUV operators is solicited to make final decisions. Algorithm 3 shows the process of HMER-SAC for AUV path planning.

4. Simulation and results

In this section, we initially outline the simulation parameters before showcasing the effectiveness and dependability of the proposed method in addressing the AUV path planning problem. This simulation encompasses both the training and testing processes. Comparative analysis against the baseline algorithm highlights the heightened suitability of the proposed method in complex environments. Lastly, the effects of key parameter are explored through comparative experiments.

Table 1
Parameters of the simulation environment.

Parameter	Meaning	Value
$L_{Env} \times W_{Env} \times D_{Env}$	Size of the simulation environment	10000 m $\times$ 10000 m $\times$ (-500) m
σ, μ	Terrain-detectable range in forward and lateral directions	20 m, 20 m
$N_\sigma \times N_\mu$	Size of the GEM data from terrain scanning device	5 $\times$ 5
b	Diffusivity parameter for Fermi function	0.5
γ_c	Proportional constant for ocean currents	0.01
d_{max}	Maximum threat areas perception radius	25 m
H_s	Safe distance to the seafloor	10 m
k_G, k_u, k_v, k_w	Normalization constants for state space	50, 1, 1, 1
$\epsilon_u, \epsilon_v, \epsilon_w$	Linear velocities for action space	1.5, 1.5, 1.5
r_s	Success reward	120
r_f	Failure punishment	-100
$\lambda_1, \lambda_2, \lambda_3, \lambda_4$	Weights of the sub-reward functions	0.45, 0.2, 0.2, 0.15

Table 2
Hyperparameters of the HMER-SAC.

Hyperparameter	Meaning	Value
$\mathcal{W}_{CA}, \mathcal{W}_{DA}, \mathcal{W}_{OC}$	The structure of policy network and action-state value networks for the Collision Avoidance Network, Detection Avoidance Network and Ocean Currents Utilization Network	30 $\times$ 256 $\times$ 256 $\times$ 11, 12 $\times$ 256 $\times$ 256 $\times$ 11, 8 $\times$ 256 $\times$ 256 $\times$ 11
$\mathcal{W}_{DN}$	The structure of policy network and action-state value networks for the Decision-Maker Network	40 $\times$ 256 $\times$ 256 $\times$ 3
$\lambda_\theta, \lambda_\phi, \lambda_a$	Learning rate	0.0001
Optimizer	The optimization method for training soft Q-network, policy network and temperature parameter	Adam (Kingma and Ba, 2014)
γ	Discount factor	0.99
τ	Soft update rate for target networks	0.01
B_P, B_E	Prior experience buffer and exploitation experience buffer	50000, 50000
N_B	Batch size	256
k_ξ	Constant parameter for MER	0.001

4.1. Simulation settings

To ensure a highly accurate simulation of the underwater operational mission for in a complex environment, we construct a simulation environment measuring 10000 m $\times$ 10000 m $\times$ (-500) m for training and testing of the proposed method. The simulation environment comprises 3 sets of GEM terrain maps, each randomly integrating 10 sets of threat areas and ocean currents, yielding a total of 30 distinct simulation scenarios. Specifically, 80% of these scenarios are allocated for the training process, while the remaining 20% are earmarked for testing purposes. The primary parameters defining the simulation environment are detailed in Table 1.

The main hyperparameters for the HMER-SAC are listed in Table 2 below. In order to demonstrate the effectiveness of the proposed method, the SAC, the PER-SAC, and the HSAC are used as comparison algorithms. Wherein, the PER-SAC is the SAC algorithm combined with Prioritized Experience Replay (PER) (Schaul et al., 2015), a mechanism for sampling based on sample priority. The HSAC is a without MER version of the proposed method. Moreover, the hidden layer structure and other hyperparameters of the networks are consistent with the HMER-SAC. All simulations are performed on a computer equipped with the Intel Core i7-12700H CPU, NVIDIA RTX 3060 GPU, and Python.

4.2. Training process and results

The initial step in addressing the AUV path planning problem using deep reinforcement learning involves training the agents. Each algorithm is trained in the simulation environment lasting 15,000 episodes. For each episode, the environment is randomly chosen from the above 40 training scenarios with random initial and target positions.

In order to ascertain the convergence and stability of the proposed method, an analysis of the algorithm's performance during the training process is conducted, with a focus on the average reward (the mean of accumulated rewards in the latest 100 episodes) and success rate (the percentage of the AUV successfully reaching the target position in

Table 3
KPIs of the algorithms in the training process.

Algorithm	CT↓	SC↓	SRC (%)↑	AP (%)↑	ARC↑
SAC	9688	1.86	82.94	86.51	1631.25
PER-SAC	8551	1.39	83.72	88.03	1631.78
HSAC	8670	1.37	86.27	88.71	1665.15
HMES-SAC	5475	0.96	87.63	89.25	1677.01

the latest 100 episodes). Five repetitions of the training experiments are conducted using different random seeds and the average is subsequently calculated to circumvent the potential influence of random numbers. To facilitate a quantitative comparison of the results in the training process, the following key performance indicators (KPIs) are used to compare the algorithms in detail (Zijian et al., 2023).

- (1) Convergence Time (CT): the number of episodes where the success rate reaches 70%.
- (2) Stability after Convergence (SC): the standard deviation of the success rate in the last 1000 episodes of the training process.
- (3) Success Rate after Convergence (SRC): the average success rate in the last 1000 episodes of the training process.
- (4) Algorithm Peak (AP): the highest success rate in the training process.
- (5) Average Reward after Convergence (ARC): the average of the episodic average reward in the last 1000 episodes of the training process.

The results of the algorithms including average reward and success rate in the training process are shown in Fig. 8, and the KPIs are calculated and listed in Table 3. In Fig. 8, the solid line represents the statistical mean of the average reward and success rate across the five training runs, and shaded areas of matching colors illustrate the upper and lower bounds of the statistical value.

The average reward curve in Fig. 8(a) illustrates the gradual increase in the average reward obtained by each algorithm during the training process. The average reward starts from an initial lower value

Table 4
Environmental parameters of the test scenarios.

Scenario	Start position p_0 (m)	Target position p_T (m)	Sonobuoys		Ocean Currents		
			p_s (m)	R_0 (m)	p_c (m)	Γ_c (m ² /s)	δ_c (m)
1	(500,500,-395)	(9500,9500,-350)	(6500,7000,-50)	1500	(-1000,-1000,-100)	5000	500
			(2100,2900,-50)		(13000,13000,-100)	5000	300
			(6000,3000,-50)		(0,-10000,-100)	-5000	200
2	(500,9500,-190)	(9500,500,-470)	(2250,7000,-50)	1500	(-2000,3000,-100)	5000	100
			(5750,6000,-50)		(12000,12000,-100)	5000	200
			(6750,2500,-50)		(3000,-2000,-100)	-7500	100
3	(9500,9500,-285)	(500,500,-400)	(2750,6500,-50)	1500	(-9000,6500,-100)	7000	200
			(6500, 6500,-50)		(-5000,12500,-100)	7000	300
			(3750,3250,-50)		(13000,13000,-300)	2500	300
4	(500,9500,-350)	(9500,500,-300)	(2750,6500,-50)	1500	(13000,0,-100)	-2000	400
			(6500,6500,-50)		(-5000,3000,-100)	-2000	150
			(6250,2750,-50)		(0,12000,-100)	4000	600
5	(500,500,-430)	(9500,9500,-370)	(7000,3350,-50)	1500	(3000,0,-100)	-3000	300
			(6500,6500,-50)		(13000,13000,-100)	4000	300
			(3750,3250,-50)		(12000,1500,-400)	-6000	300
6	(500,9500,-230)	(9500,500,-395)	(6500,3000,-50)	1500	(-100,14000,-100)	9000	600
			(6500,6500,-50)		(-1000,1000,-200)	6000	450
			(3250,6250,-50)		(12000,1500,-400)	6000	400
					(-100,14000,-100)	9000	800
					(-1000,1000,-200)	-6000	600

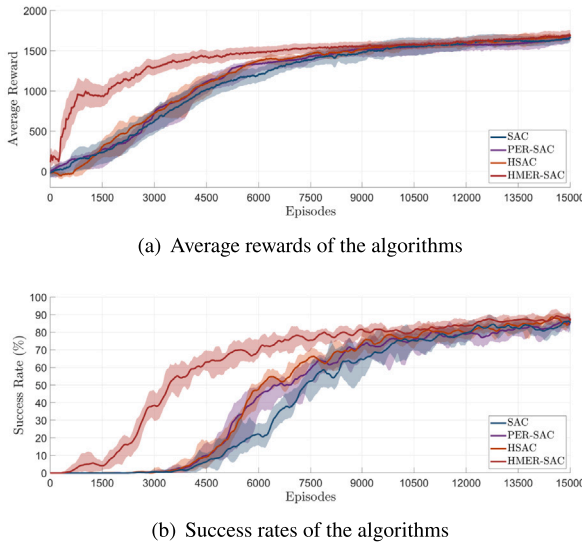


Fig. 8. Results of the algorithms in the training process.

and converges near 1600 to 1700. Specifically, according to the CT in Table 3, the proposed HMER-SAC only needs 5475 episodes to reach convergence, which is the fastest among all algorithms. Compared with the SAC, the PER-SAC, and the HSAC, the convergence speed of the HMER-SAC improves by 43.49%, 35.97%, and 36.85% respectively.

The SC and the SRC reflect the stability and effectiveness of the trained algorithm, respectively. A smaller SC means better convergence stability and a larger SRC means a higher success probability in application. As shown in Table 3, the HMER-SAC has the smallest SC (0.96), which is better than HSAC(1.37), PER-SAC(1.39) and SAC(1.86). Similarly, HMER-SAC has the best performance on the indicator SRC, improving 5.65%, 4.67%, 1.57% compared to the SAC, the PER-SAC, and the HSAC.

Furthermore, the AP delineates the upper limit of the policy throughout the entirety of the training process, whereas the ARC represents the average level of the policy after convergence. In the comparison of these two KPIs, the proposed HMER-SAC still presents better performance than other algorithms.

From the above results, it can be seen that compared with the basic SAC algorithm, the performance of HSAC in the training process has a

small improvement, which shows the effectiveness of the hierarchical learning strategy. On this basis, combined with MER, the proposed HMER-SAC has more excellent performance in terms of convergence and stability because of the prior experiences.

4.3. Testing process and results

To assess the validity of the training results, the algorithms are tested across 6 distinct scenarios. Three KPIs, comprising Path Length (PL), Average Safety Indicator (ASI), and Average Current Utilization Indicator (ACUI), are computed and compared in the testing process. PL refers to the sum of the Euclidean distances between adjacent waypoints defined as Eq. (36). ASI represents the average weighted sum of collision avoidance reward and detection avoidance reward obtained per time-step, as defined in Eq. (37). ACUI denotes the average ocean currents utilization reward per time-step, as defined in Eq. (38). Considering the requirements of the AUV path planning problem in Eq. (1), it is preferable for PL to be minimized, while ASI and ACUI are desired to be maximized.

$$PL = \sum_{i=1}^{n-1} \|p_i - p_{i+1}\|_2 \quad (36)$$

$$ASI = \frac{1}{n} \sum_{i=1}^n \frac{\lambda_2 r_{2,i} + \lambda_3 r_{3,i}}{\lambda_2 + \lambda_3} \quad (37)$$

$$ACUI = \frac{1}{n} \sum_{i=1}^n r_{4,i} \quad (38)$$

The path planning results of the algorithms in different simulation scenarios are depicted in Fig. 9. Table 4 lists the environmental parameters for these test scenarios, including the coordinates of the start and target positions and the parameter settings for the sonobuoys and ocean currents. In Fig. 9, the three-dimensional surface represents the underwater terrain, the red areas represent the detection areas of the sonobuoys with a radius of R_0 , the blue arrows represent ocean currents, the black circle and square represent the starting position and target position respectively, and the curves in different colors represent the path results of the algorithms. The KPIs of the algorithms in different scenarios for the testing process are calculated and listed in Table 5.

Fig. 9 illustrates that both the proposed HMER-SAC and the compared algorithms can achieve path planning in each simulation scenario. This result is consistent with the result in the training process that the SRC of each algorithm is above 80%. The result in Table 5

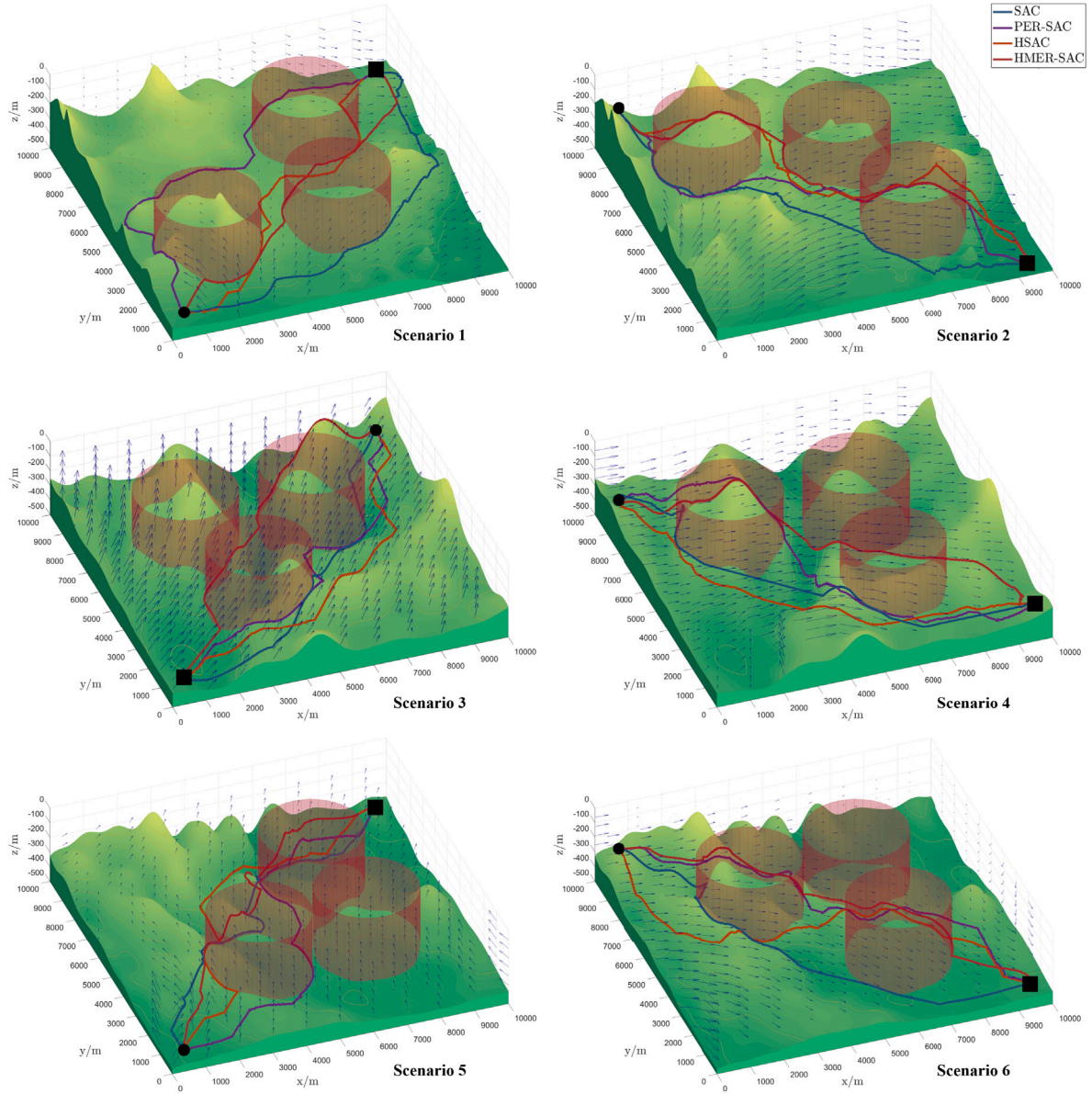


Fig. 9. Path planning results of the algorithms in different simulation scenarios.

shows that the proposed HMER-SAC in this paper outperforms the compared algorithm in most of the KPIs, and only slightly underperforms the HSAC algorithm in the ACUI indicator in Scenario 5. This indicates that the HRL and MER adopted in the proposed method can effectively combine and efficiently learn from prior experiences and exploration experiences in different scenarios. At the same time, due to the uncertainty associated with complex environments and the randomness of the algorithms, there exists a small possibility that the proposed method performs worse than the comparison algorithms in some KPIs. However, it does not detract from the superior performance of the proposed method.

Furthermore, we calculate the average value of KPIs in the scenarios for the algorithms, as shown in Table 6. The average value of KPIs in the scenarios reflects the overall level of the algorithm. Obviously, the proposed HMER-SAC presents the best results on each KPI. For PL, the proposed HMER-SAC improves by 5.41%, 7.11%, and 6.43% compared with SAC, PER-SAC, and HSAC respectively. For ASI, the improvements are 7.85%, 5.85% and 4.78%, whereas 100.80%, 19.52% and 20.10% for ACUI. This result suggests that the proposed method can obtain a better policy with the support of HRL and MER, and therefore exhibits

better performance. In addition, it is worth noting that HSAC and PER-SAC outperform basic SAC in ASI and ACUI, but perform the opposite in PL. This means that the policy learning of PER-SAC and HSAC does not ensure improvement in all measures and its adaptation to different scenarios is poorer than the proposed methods.

From the experimental results of the training and testing processes, we observe that the proposed HMER-SAC algorithm outperforms the compared algorithms. This improvement is attributed to the combined use of the HER and MER strategies. The HER strategy effectively decomposes the high-dimensional complex problem into simpler sub-problems, while MER leverages prior knowledge to accelerate agent training further. Unlike comparative algorithms, which ignorantly explore at the beginning of training, HMER-SAC obtains prior knowledge to support the training. The strategies provide more opportunities and time to obtain more and better training samples through exploration, thereby enhancing training efficiency and testing performance. As a result, the proposed HMER-SAC algorithm demonstrates superior performance in KPIs in both the training and testing process. Additionally, we believe that further enhancing the relevance of the prior knowledge

Table 5

KPIs of the algorithms in the test scenarios.

Scenario	Algorithm	PL (m) ↓	ASI ↑	ACUI ↑
1	SAC	16 638	1.60	0.01
	PER-SAC	16 933	1.69	0.27
	HSAC	14 809	1.72	0.25
	HMER-SAC	14 411	1.84	0.30
2	SAC	16 311	1.89	0.22
	PER-SAC	16 307	1.92	0.60
	HSAC	16 436	1.84	0.50
	HMER-SAC	15 622	1.97	0.64
3	SAC	14 765	1.59	−0.88
	PER-SAC	15 367	1.63	−0.76
	HSAC	15 120	1.69	−0.78
	HMER-SAC	14 374	1.77	−0.74
4	SAC	15 147	1.86	0.57
	PER-SAC	15 046	1.82	0.58
	HSAC	16 144	1.87	0.68
	HMER-SAC	14 547	1.95	0.78
5	SAC	15 405	1.80	0.72
	PER-SAC	16 499	1.87	0.65
	HSAC	15 644	1.89	0.74
	HMER-SAC	14 756	1.92	0.72
6	SAC	15 055	1.83	0.61
	PER-SAC	14 882	1.84	0.76
	HSAC	16 188	1.87	0.71
	HMER-SAC	14 566	1.95	0.81

Table 6

Average KPIs of the algorithms in the training process.

Algorithm	PL (m) ↓	ASI ↑	ACUI ↑
SAC	15 553.50	1.762	0.208
PER-SAC	15 839.00	1.795	0.350
HSAC	15 723.50	1.813	0.348
HMER-SAC	14 712.67	1.900	0.418

Table 7KPIs in the training process of different k_ξ values.

k_ξ	CT↓	SC↓	SRC(%)↑	AP(%)↑	ARC↑
0.0001	6977	1.05	86.31	88.56	1659.49
0.0005	7013	1.15	86.78	87.62	1641.45
0.001	5475	0.96	87.63	89.25	1677.01
0.005	6639	1.13	86.83	87.92	1655.35

to the actual situation during the exploration can improve the performance of the proposed method and will focus on this in subsequent studies.

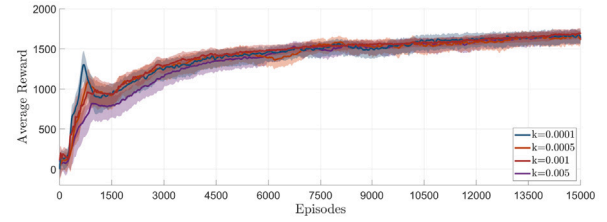
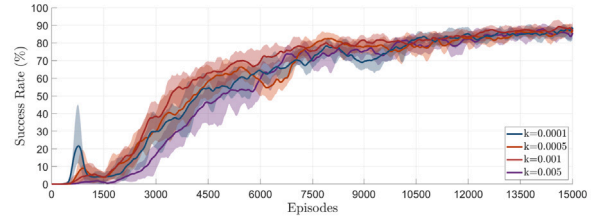
In summary, from the training and testing processes, the proposed HMER-SAC is an effective method for the AUV path planning problem in complex marine environments. It benefits from the hierarchical reinforcement learning strategy and the mixed experience replay strategy, which result in faster training convergence, improved stability, and superior performance compared to the SAC, PER-SAC, and HSAC.

4.4. Parameter effects analysis experiment

Methods based on deep reinforcement learning are sensitive to parameter changes, and appropriate parameter settings can improve the performance of the method. The HMER-SAC algorithm proposed in this paper introduces the parameter k_ξ , so it is necessary to further analyze the impact of the value of this parameter on the algorithm. In this section, we trained the HMER-SAC algorithm with different k_ξ values five times and calculated the average value of each KPI used in Section 4.2.

The results of the k_ξ values including average reward and success rate in the training process are shown in Fig. 10, and the KPIs are calculated and listed in Table 7.

In the proposed MER strategy, the ξ in Eq. (32) is used to represent the proportion of prior experiences in the training samples. At the

(a) Average rewards of the different k_ξ value(b) Success rates of the different k_ξ value**Fig. 10.** Results of the different k_ξ value in the training process.

beginning of training, this ratio is expected to be a high value, due to the low quality of the initial exploration experience and the need for the agent to upgrade its policy by learning more prior experience. As training progresses, this proportion should gradually decrease, allowing more exploration experience to be incorporated into the policy to learn a more appropriate policy for the application scenarios.

The parameter k_ξ is used to control the decrease rate of the proportional parameter ξ . The larger the value of k_ξ , the faster the percentage of the prior experience decreases. When the value of k_ξ is excessive, the agent is unable to sufficiently learn from prior experience. Conversely, when the value of k_ξ is insufficient, the agent is unable to fully leverage the policy generated from interaction. Therefore, the parameter k_ξ has a direct impact on the performance of the agent in the training process. In order to find the most appropriate value of k_ξ , we conducted four sets of experiments as shown in Fig. 10 and Table 7, and the results show that the KPIs in the training process are optimal when $k_\xi = 0.001$.

5. Conclusion

In this paper, we propose the HMER-SAC method for AUV local path planning in complex environments. Firstly, for the complex scenarios of AUV performing underwater operational missions, the simulation environment including underwater terrain, sonobuoy detection, and ocean currents is constructed, and corresponding state space, action space, and reward function are designed. Subsequently, the HMER-SAC method for AUV path planning is proposed by combining the SAC algorithm, the hierarchical reinforcement learning strategy, and the mixed experience replay strategy. The detailed computation process and implementation structure are presented. Lastly, simulation experiments show that the proposed method can effectively solve the AUV path planning problem in complex marine environments and outperform the compared algorithms in key performance indicators.

In this work, we address the AUV path planning problem in a static environment. However, in real-world scenarios, the environment is dynamic: underwater terrain obstacles may change due to various factors, sonobuoys drift with ocean currents, and ocean currents change unpredictably in strength and direction. These dynamic factors may limit the application of the proposed method. In the future, we will focus on optimizing the limitations of the proposed method to enhance its applicability.

CRediT authorship contribution statement

An Zhang: Funding acquisition, Formal analysis, Data curation.
Weixiang Wang: Writing – review & editing, Writing – original draft, Methodology.
Wenhao Bi: Formal analysis, Conceptualization.
Zhan-jun Huang: Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Wenhao Bi reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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